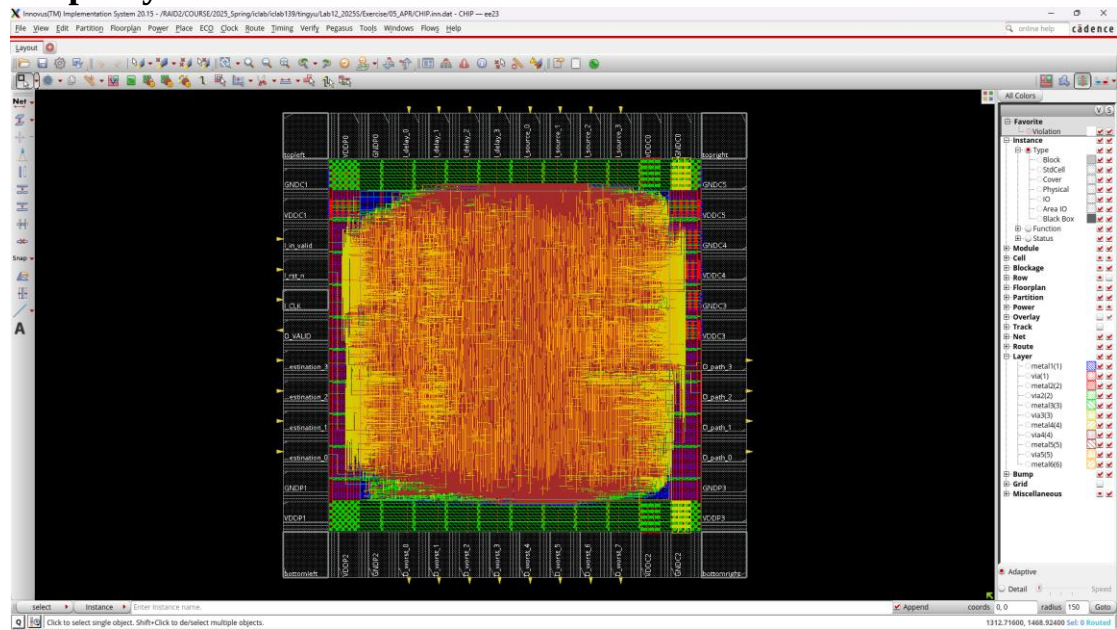
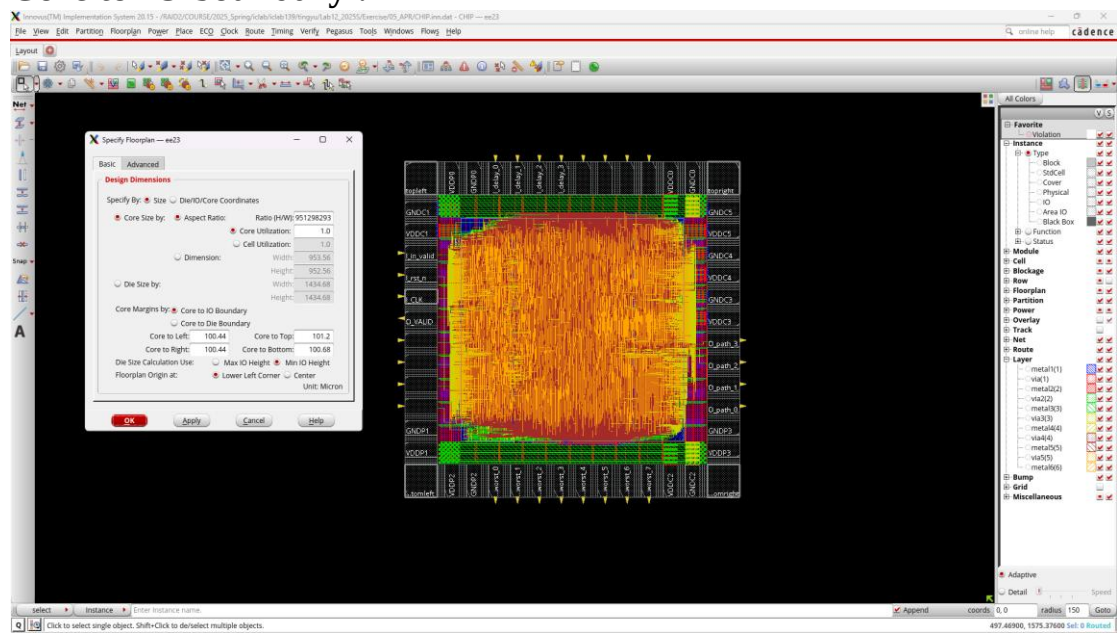


Report

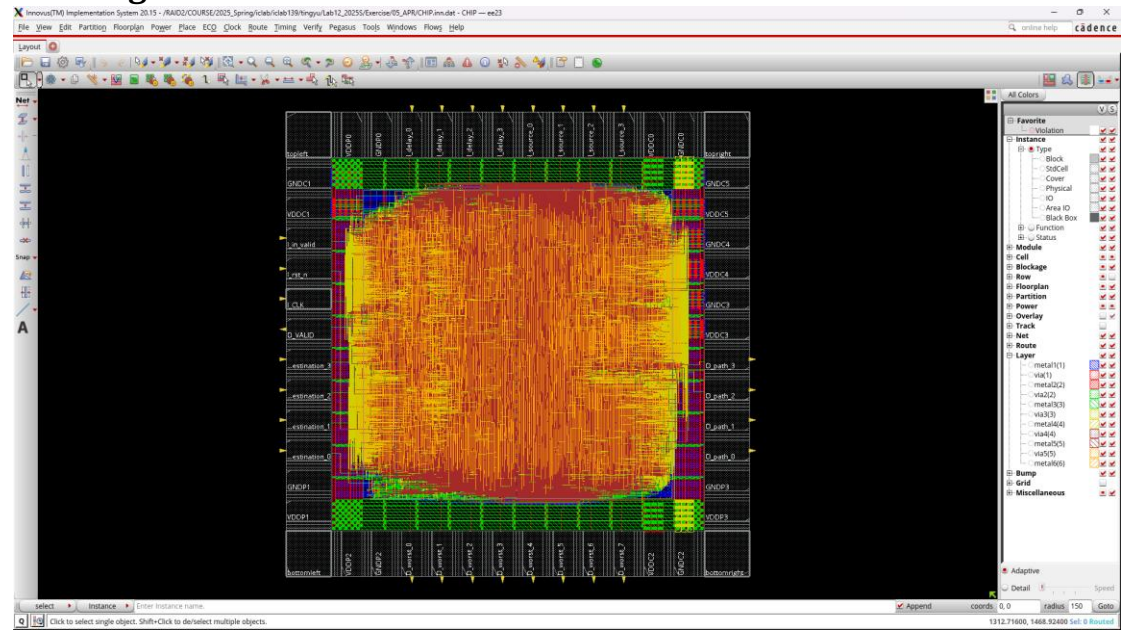
1. Chip Layout View :



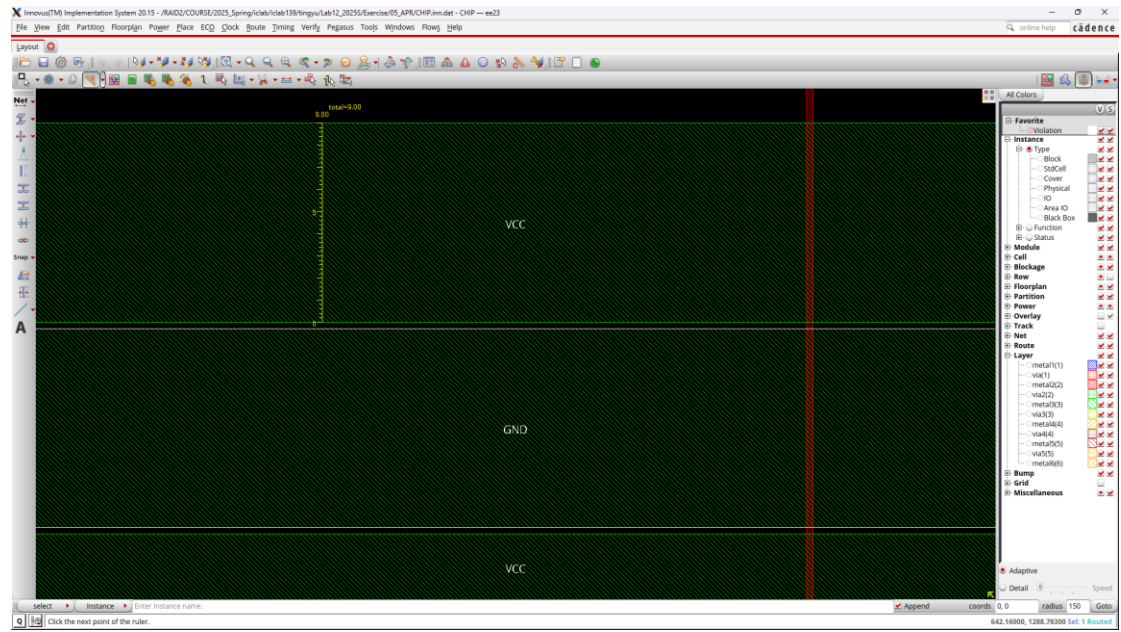
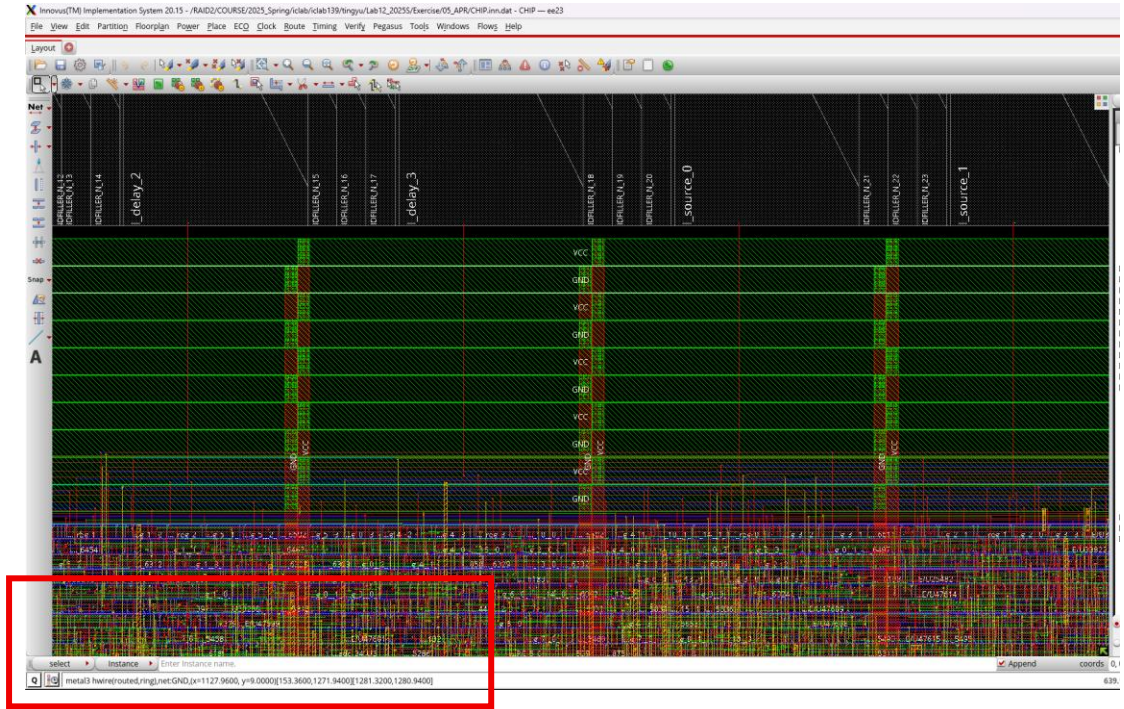
2. Core to IO boundary :



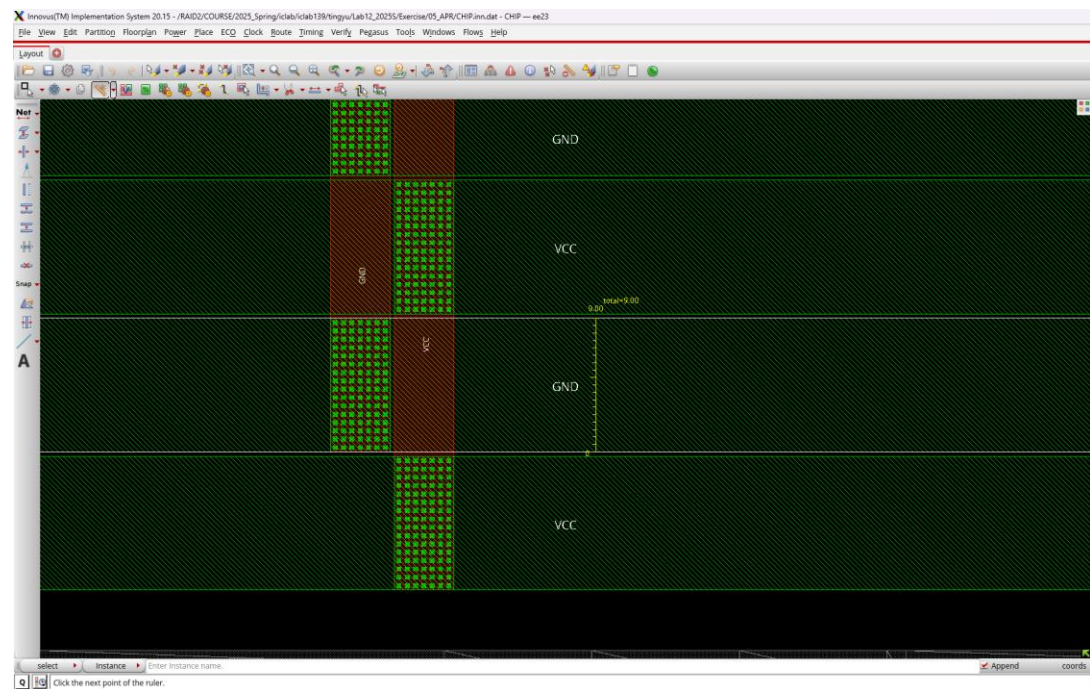
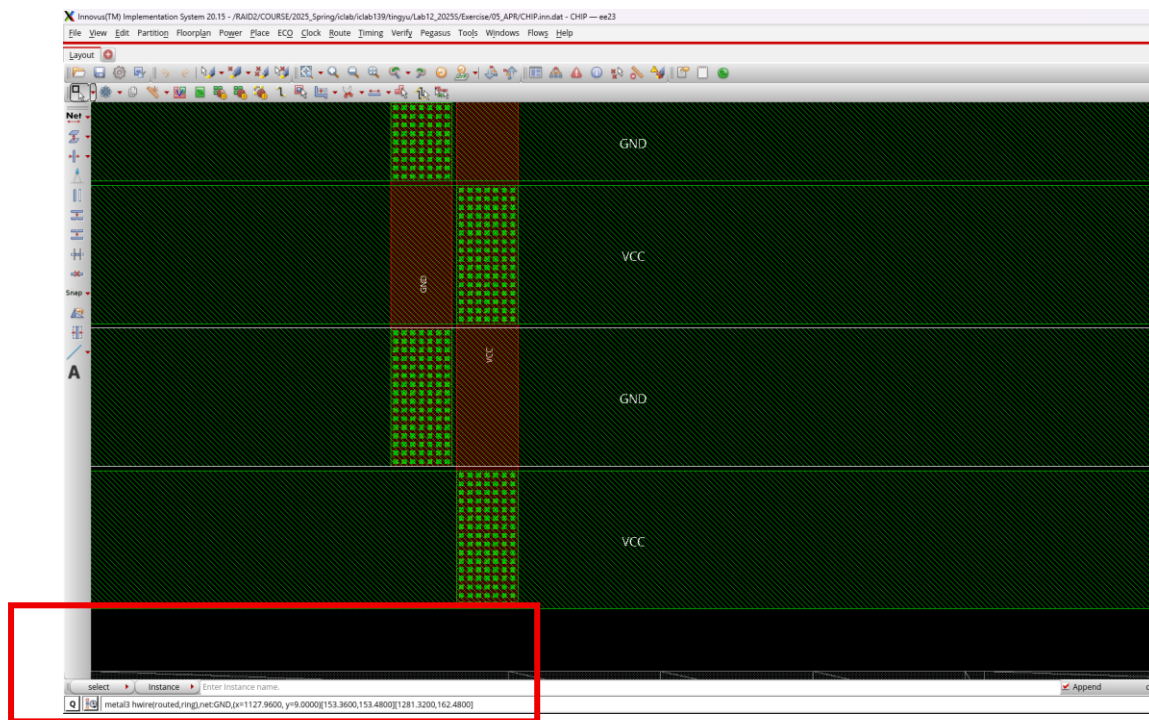
All Ring



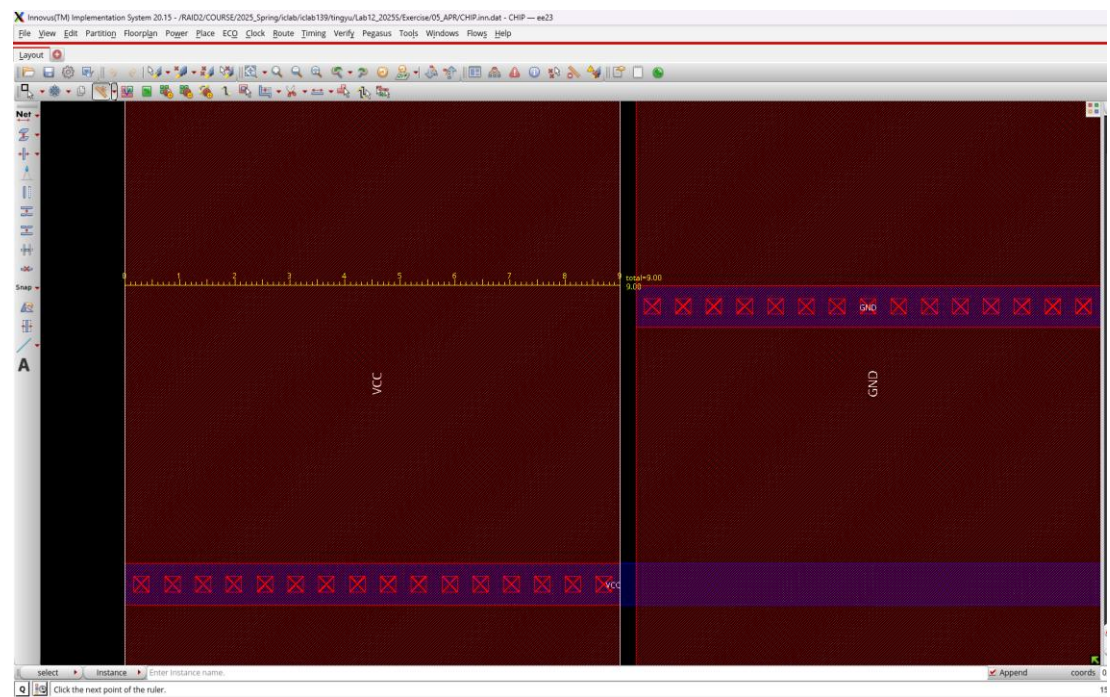
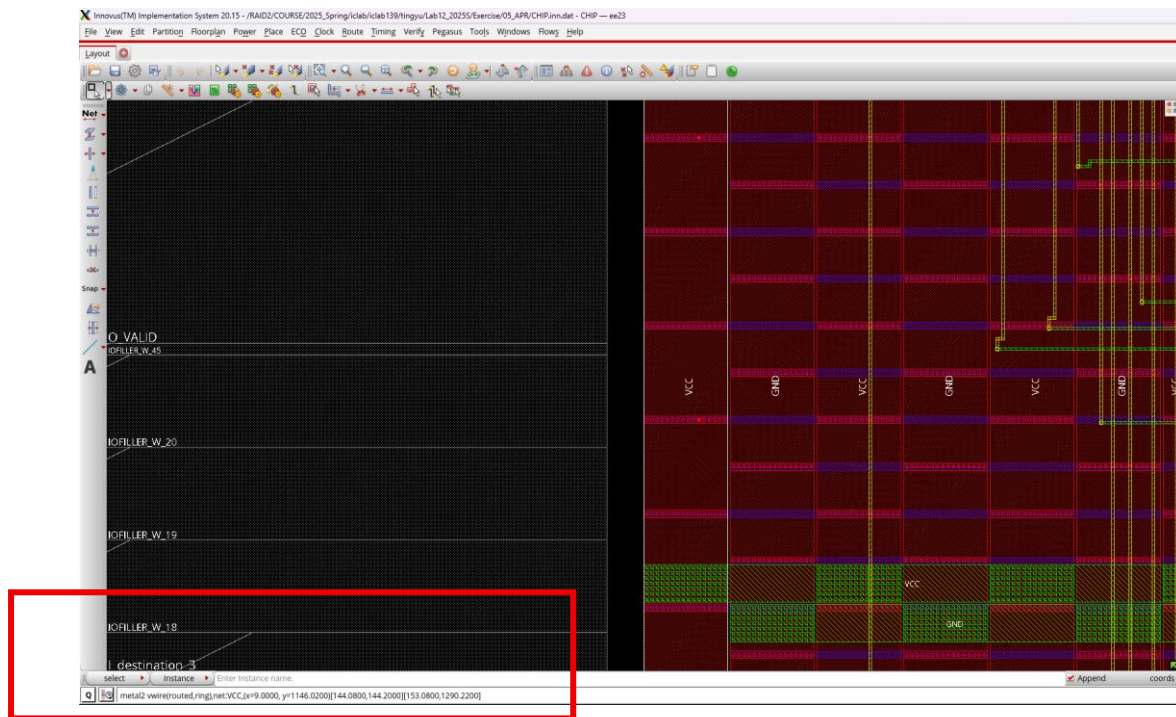
Top



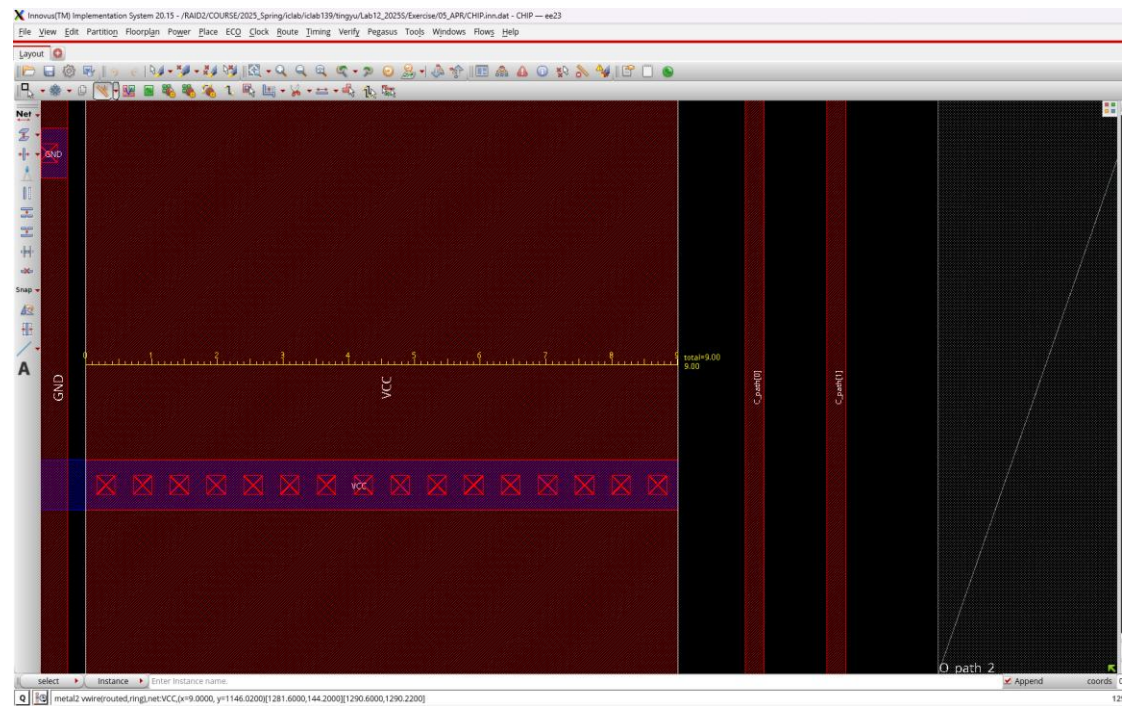
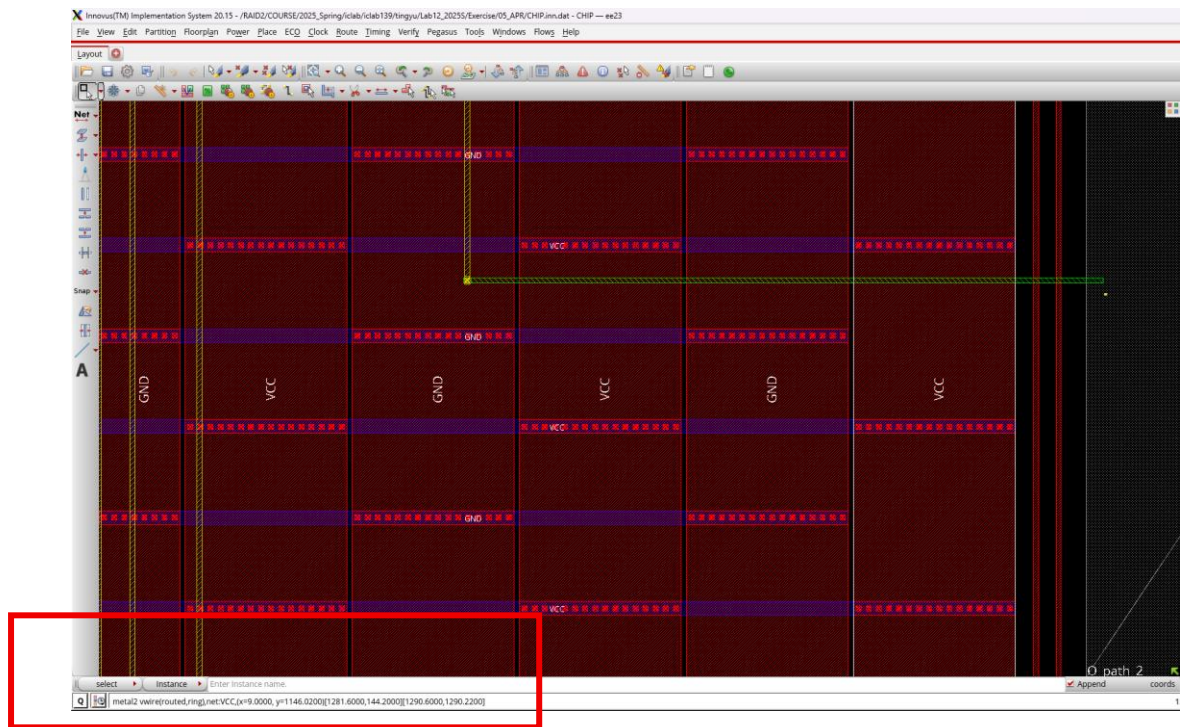
Bottom



Left



Right



4. Post-Route setup time analysis :

```
Total number of fetched objects 26209
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 26165, 100.0 percent of the nets selected for SI analysis
End delay calculation. (MEM=3717.11 CPU=0:00:08.6 REAL=0:00:09.0)
End delay calculation (fullDC). (MEM=3717.11 CPU=0:00:09.1 REAL=0:00:09.0)
Loading CTE timing window with twFlowType 0...(CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 3717.1M)
Add other clocks and setupCteToAEClockMapping during iter 1
Loading CTE timing window is completed (CPU = 0:00:00.2, REAL = 0:00:00.0, MEM = 3717.1M)
Starting SI iteration 2
Start delay calculation (fullDC) (1 T). (MEM=3694.23)
Glitch Analysis: View av_func_mode_max -- Total Number of Nets Skipped = 0.
Glitch Analysis: View av_func_mode_max -- Total Number of Nets Analyzed = 26209.
Total number of fetched objects 26209
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 26165, 14.6 percent of the nets selected for SI analysis
End delay calculation. (MEM=3734.4 CPU=0:00:04.0 REAL=0:00:04.0)
End delay calculation (fullDC). (MEM=3734.4 CPU=0:00:04.1 REAL=0:00:04.0)
*** Done Building Timing Graph (cpu=0:00:15.5 real=0:00:15.0 totSessionCpu=0:07:17 mem=3734.4M)

-----
timeDesign Summary
-----

Setup views included:
av_func_mode_max

-----
| Setup mode | all | regreg | default |
-----+-----+-----+-----+
| WNS (ns): | 0.075 | 0.075 | 0.223 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 1415 | 688 | 727 |
-----

-----
| | Real | Total |
-----+-----+-----+
| DRVs | Nr nets(terms) | Worst Vio | Nr nets(terms) |
-----+-----+-----+
| max_cap | 0 (0) | 0.000 | 0 (0) |
| max_tran | 0 (0) | 0.000 | 0 (0) |
| max_fanout | 0 (0) | 0 | 0 (0) |
| max_length | 0 (0) | 0 | 0 (0) |
-----

Density: 61.795%
(125.795% with Fillers)
Total number of glitch violations: 0
Reported timing to dir timingReports
Total CPU time: 18.35 sec
Total Real time: 20.0 sec
Total Memory Usage: 3711.671875 Mbytes
Reset AAE Options
*** timeDesign #2 [finish] : cpu/real = 0:00:18.3/0:00:20.0 (0.9), totSession cpu/real = 0:07:18.5/0:49:01.9 (0.1), mem = 3711.7M
Unnovus i>

Loading remote monitoring, please wait...
here: https://mobaxterm.mobatek.net
```

5. Post-Route hold time analysis :

```
Starting SI iteration 1 using Infinite Timing Windows
#####
# Design Stage: PostRoute
# Design Name: CHIP
# Design Mode: 180nm
# Analysis Mode: MMCM OCV
# Parasitics Mode: SPEF/RCDB
# Signoff Settings: SI On
#####
AAE INFO: 1 threads acquired from CTE.
Start delay calculation (fullDC) (1 T). (MEM=3640.02)
*** Calculating scaling factor for lib_min libraries using the default operating condition of each library.
Total number of fetched objects 26209
AAE INFO-618: Total number of nets in the design is 26165, 100.0 percent of the nets selected for SI analysis
End delay calculation. (MEM=3700.33 CPU=0:00:08.8 REAL=0:00:09.0)
End delay calculation (fullDC). (MEM=3700.33 CPU=0:00:09.3 REAL=0:00:09.0)
Loading CTE timing window with twFlowType 0...(CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 3700.3M)
Add other clocks and setupCteToAEClockMapping during iter 1
Loading CTE timing window is completed (CPU = 0:00:00.1, REAL = 0:00:00.0, MEM = 3700.3M)
Starting SI iteration 2
Start delay calculation (fullDC) (1 T). (MEM=3676.45)
Glitch Analysis: View av_func_mode_min -- Total Number of Nets Skipped = 0.
Glitch Analysis: View av_func_mode_min -- Total Number of Nets Analyzed = 26209.
Total number of fetched objects 26209
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 26165, 0.0 percent of the nets selected for SI analysis
End delay calculation. (MEM=3716.62 CPU=0:00:00.2 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=3716.62 CPU=0:00:00.2 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:11.1 real=0:00:11.0 totSessionCpu=0:06:57 mem=3716.6M)

-----
timeDesign Summary
-----

Hold views included:
av_func_mode_min

-----
| Hold mode | all | reg2reg | default |
-----+-----+-----+
| WNS (ns): | 0.263 | 0.263 | 5.124 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 1415 | 688 | 727 |
-----

Density: 61.795%
(125.795% with Fillers)
Reported timing to dir timingReports
Total CPU time: 13.33 sec
Total Real time: 14.0 sec
Total Memory Usage: 3640.882812 Mbytes
Reset AAE Options
*** timeDesign #1 [finish] : cpu/real = 0:00:13.3/0:00:13.7 (1.0), totSession cpu/real = 0:06:57.4/0:48:04.2 (0.1), mem = 3640.9M
Unnovus i>

Loading remote monitoring, please wait...
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```


6. DRC result :

```
VERIFY DRC ..... Sub-Area : 11 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1209.600 241.920 1434.680 483.840} 12 of 36
VERIFY DRC ..... Sub-Area : 12 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 483.840 241.920 725.760} 13 of 36
VERIFY DRC ..... Sub-Area : 13 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {241.920 483.840 483.840 725.760} 14 of 36
VERIFY DRC ..... Sub-Area : 14 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {483.840 483.840 725.760 725.760} 15 of 36
VERIFY DRC ..... Sub-Area : 15 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {725.760 483.840 967.680 725.760} 16 of 36
VERIFY DRC ..... Sub-Area : 16 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {967.680 483.840 1209.600 725.760} 17 of 36
VERIFY DRC ..... Sub-Area : 17 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1209.600 483.840 1434.680 725.760} 18 of 36
VERIFY DRC ..... Sub-Area : 18 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 725.760 241.920 967.680} 19 of 36
VERIFY DRC ..... Sub-Area : 19 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {241.920 725.760 483.840 967.680} 20 of 36
VERIFY DRC ..... Sub-Area : 20 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {483.840 725.760 725.760 967.680} 21 of 36
VERIFY DRC ..... Sub-Area : 21 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {725.760 725.760 967.680 967.680} 22 of 36
VERIFY DRC ..... Sub-Area : 22 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {967.680 725.760 1209.600 967.680} 23 of 36
VERIFY DRC ..... Sub-Area : 23 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1209.600 725.760 1434.680 967.680} 24 of 36
VERIFY DRC ..... Sub-Area: {0.000 967.680 241.920 1209.600} 25 of 36
VERIFY DRC ..... Sub-Area : 25 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {241.920 967.680 483.840 1209.600} 26 of 36
VERIFY DRC ..... Sub-Area : 26 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {483.840 967.680 725.760 1209.600} 27 of 36
VERIFY DRC ..... Sub-Area : 27 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {725.760 967.680 967.680 1209.600} 28 of 36
VERIFY DRC ..... Sub-Area : 28 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {967.680 967.680 1209.600 1209.600} 29 of 36
VERIFY DRC ..... Sub-Area : 29 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1209.600 967.680 1434.680 1209.600} 30 of 36
VERIFY DRC ..... Sub-Area : 30 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 1209.600 241.920 1434.680} 31 of 36
VERIFY DRC ..... Sub-Area : 31 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {241.920 1209.600 483.840 1434.680} 32 of 36
VERIFY DRC ..... Sub-Area : 32 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {483.840 1209.600 725.760 1434.680} 33 of 36
VERIFY DRC ..... Sub-Area : 33 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {725.760 1209.600 967.680 1434.680} 34 of 36
VERIFY DRC ..... Sub-Area : 34 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {967.680 1209.600 1209.600 1434.680} 35 of 36
VERIFY DRC ..... Sub-Area : 35 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1209.600 1209.600 1434.680 1434.680} 36 of 36
VERIFY DRC ..... Sub-Area : 36 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:04.7 ELAPSED TIME: 5.00 MEM: 41.0M) ***

innovus 1> █
```

啟用 Windows
移至 [設定] 以啟用 Windows +

Loading remote monitoring, please wait...

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7. LVS result :

```
Type 'man IMPCTE-291' for more detail.
**WARN: (IMPCTE-291): FILLER_T_5_4642 is a physical only instance and is ignored. This indicates that the instance is physical only (e.g. a filler/decap cell) and is in the database, but not in timing graph since physical cells do not have timing information. Verify the instance is a physical only cell to confirm that this warning can be ignored.
Type 'man IMPCTE-291' for more detail.
**WARN: (IMPCTE-291): FILLER_T_5_4655 is a physical only instance and is ignored. This indicates that the instance is physical only (e.g. a filler/decap cell) and is in the database, but not in timing graph since physical cells do not have timing information. Verify the instance is a physical only cell to confirm that this warning can be ignored.
Type 'man IMPCTE-291' for more detail.
**WARN: (IMPCTE-291): FILLER_T_5_4657 is a physical only instance and is ignored. This indicates that the instance is physical only (e.g. a filler/decap cell) and is in the database, but not in timing graph since physical cells do not have timing information. Verify the instance is a physical only cell to confirm that this warning can be ignored.
Type 'man IMPCTE-291' for more detail.
**WARN: (IMPCTE-291): FILLER_T_5_4627 is a physical only instance and is ignored. This indicates that the instance is physical only (e.g. a filler/decap cell) and is in the database, but not in timing graph since physical cells do not have timing information. Verify the instance is a physical only cell to confirm that this warning can be ignored.
Type 'man IMPCTE-291' for more detail.
**WARN: (IMPCTE-291): FILLER_T_5_4632 is a physical only instance and is ignored. This indicates that the instance is physical only (e.g. a filler/decap cell) and is in the database, but not in timing graph since physical cells do not have timing information. Verify the instance is a physical only cell to confirm that this warning can be ignored.
Type 'man IMPCTE-291' for more detail.
**WARN: (IMPCTE-291): FILLER_T_5_4834 is a physical only instance and is ignored. This indicates that the instance is physical only (e.g. a filler/decap cell) and is in the database, but not in timing graph since physical cells do not have timing information. Verify the instance is a physical only cell to confirm that this warning can be ignored.
Type 'man IMPCTE-291' for more detail.
**WARN: (IMPCTE-291): FILLER_T_5_4855 is a physical only instance and is ignored. This indicates that the instance is physical only (e.g. a filler/decap cell) and is in the database, but not in timing graph since physical cells do not have timing information. Verify the instance is a physical only cell to confirm that this warning can be ignored.
Type 'man IMPCTE-291' for more detail.
**WARN: (EMS-27): Message (IMPCTE-291) has exceeded the current message display limit of 20.
To increase the message display limit, refer to the product command reference manual.
innovus 1> set_power_rail_display -plot none

innovus 1> set_power_rail_display -plot ir

innovus 1>
innovus 1> VERIFY_CONNECTIVITY use new engine.

***** Start: VERIFY CONNECTIVITY *****
Start Time: Thu May 29 19:28:40 2025

Design Name: CHIP
Database Units: 1000
Design Boundary: (0.0000, 0.0000) (1434.6800, 1434.6800)
Error Limit = 1000; Warning Limit = 50
Check all nets
**** 19:28:41 **** Processed 5000 nets.
**** 19:28:41 **** Processed 10000 nets.
**** 19:28:41 **** Processed 15000 nets.
**** 19:28:41 **** Processed 20000 nets.
**** 19:28:41 **** Processed 25000 nets.

Begin Summary
Found no problems or warnings.
End Summary

End Time: Thu May 29 19:28:42 2025
Time Elapsed: 0:00:02.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:01.1 MEM: -0.438M)

innovus 1> █
```

啟用 Windows
移至 [設定] 以啟用 Windows +

Loading remote monitoring, please wait...

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```
2. 140.110.122.198 (u7117884) [3. e022.siz.lee.nyu.edu.tw (clab): x
```

```
** WARN: (VOLTUS_POWER-2152): Instance VDDC4 (VCCKD) has no static power.
** WARN: (VOLTUS_POWER-2152): Instance GND_C4 (GNDCKD) has no static power.
** WARN: (VOLTUS_POWER-2152): Instance VDDC5 (VCCKD) has no static power.
** WARN: (VOLTUS_POWER-2152): Instance GND_C5 (GNDCKD) has no static power.
*

Total Power
-----
Total Internal Power: 3.77009352 55.4977%
Total Switching Power: 3.80779320 44.2264%
Total Leakage Power: 0.02270623 0.3339%
Total Power: 6.80059294
-----
** WARN: (VOLTUS_POWER-2041): There are some instances in the design which are not connected to any power or ground nets.
These instances will be added to default power/ground cell. Use the
Use 'itaputil list <uti-files>' command to get the list of instances.

** WARN: (VOLTUS_POWER-3424): Cell OA13S has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell OR3S has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell AN4S has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell OR3B1S has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell A0112 has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell MOA1I1H has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell BUF3 has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell AN3 has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell A022T has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell INV12CK has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell OR2B1 has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell ND3 has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell AN2B1T has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell INV4CK has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell VCC3I00 has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell OA1112HP has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWER-3424): Cell OR3 has no power pin defined in LEF/PGV.

Loading remote monitoring, please wait...
Where: https://mobaxterm.mobatek.net
```

10. IR Drop Results :

